human

Find all buildings that are within 1km of Exeter

host_agent

Find all buildings that are within 1km of Exeter

Tool Calls

Tool: send_message

Arguments: {'target': 'planning_agent', 'task_description': 'Find all buildings that are within 1km of Exeter. Provide the general steps to solve this geospatial query.'}

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are buildings, and we need to find as many entries as possible. The condition is that these buildings must be within 1km of Exeter.

output steps: ["Find Exeter 1 area", "Find buildings in Exeter as many search results", "Apply condition buildings within 1km of Exeter"]

Tool Calls

Tool: send_message

Arguments: {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter'}

named_area_agent

Find the area polygon for Exeter

Tool Calls

Tool: call_os_ngd

Arguments: {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}

Tool Response:

Search results have been found.

Artifact generated is : exeter_area_polygon,

Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter.,

count of records fetched : 98.

Tool: send_message

Arguments: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'.

Tool: generate_metadata_for_artifacts

Arguments: {'artifact_names': ['exeter_area_polygon']}

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter using multiple columns to ensure we get the correct polygon for Exeter
    filtered = gdf[(gdf['name1_text'].str.lower() == 'exeter') |
                   (gdf['description'].str.contains('Exeter', case=False, na=False)) |
                   (gdf['theme'].str.contains('exeter', case=False, na=False))]
    # Further refine: prefer exact match in name1_text if available
    exact = filtered[filtered['name1_text'].str.lower() == 'exeter']
    if not exact.empty:
        filtered = exact
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter."
    return [summary, 'filtered_exeter_area_polygon', 'GeoDataFrame containing only the polygon(s) for Exeter, filtered using name and description columns.', fi
```

Filtered to 1 polygon for Exeter. The result is saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) for Exeter, filtered using name and description columns.

named_area_agent

The specific polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This GeoDataFrame contains only the polygon(s) for Exeter, filtered using name and description columns.

Tool Calls

Tool: send_message

Arguments: {'target': 'buildings', 'task_description': "Find all buildings in the area defined by the artifact 'filtered_exeter_area_polygon'."}

buildings_agent

Find all buildings in the area defined by the artifact 'filtered_exeter_area_polygon'.

Tool: call_os_ngd

Arguments: {'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'exeter_buildings'}

Tool Response:

Multiple search results have been found in multiple datasets. A summary of each is provided.

Artifacts generated are : ['buildings_exeter_buildings', 'buildingline_exeter_buildings', 'buildingpart_exeter_buildings'],

Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Find all buildings in the area defined by the artifact 'filtered_exeter_area_polygon'. using filters None.", "A geopandas dataframe containing building line data with filters and bbox applied for the query Find all buildings in the area defined by the artifact 'filtered_exeter_area_polygon'. using filters None.", "A geopandas dataframe containing building part data with filters and bbox applied for the query Find all buildings in the area defined by the artifact 'filtered_exeter_area_polygon'. using filters None."],

counts of records fetched are : [69169, 69342, 85480].

The search returned the following artifacts for buildings in the 'filtered_exeter_area_polygon':

1. **buildings_exeter_buildings**: A geopandas dataframe containing building data. (69,169 records)
2. **buildingline_exeter_buildings**: A geopandas dataframe containing building line data. (69,342 records)

3. **buildingpart_exeter_buildings**: A geopandas dataframe containing building part data. (85,480 records)

These artifacts contain the building information for the specified area.

Tool Calls

Tool: generate_metadata_for_all_artifacts

Arguments: {}

Tool Response:
Metadata for all artifacts generated.

Tool Calls

Tool: send_message

Arguments: {'target': 'plotting_agent', 'task_description': "Plot all buildings from 'buildings_exeter_buildings', 'buildingline_exeter_buildings', and 'buildingpart_exeter_buildings' that are within 1km of the 'filtered_exeter_area_polygon' (Exeter area)."}

plotting_agent

Plot all buildings from 'buildings_exeter_buildings', 'buildingline_exeter_buildings', and 'buildingpart_exeter_buildings' that are within 1km of the 'filtered_exeter_area_polygon' (Exeter area).

Tool: generate_metadata_for_artifacts

Arguments: {'artifact_names': ['buildings_exeter_buildings', 'buildingline_exeter_buildings', 'buildingpart_exeter_buildings', 'filtered_exeter_area_polygon']}

Tool Code Executor

```
def plot_buildings_within_1km_of_exeter_area(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import Polygon, LineString, Point
    from shapely.ops import unary_union
    import pandas as pd

    # Helper to remove all non-serializable columns
```

```python
def clean_gdf(gdf):
    for col in gdf.columns:
        if gdf[col].dtype == 'datetime64[ns]' or gdf[col].dtype.name == 'datetime64[ns, UTC]':
            gdf[col] = gdf[col].astype(str)
        elif gdf[col].dtype.name == 'object':
            # Try to convert objects that are pd.Timestamp
            gdf[col] = gdf[col].apply(lambda x: str(x) if isinstance(x, pd.Timestamp) else x)
    return gdf

# Unpack data
buildings = clean_gdf(data[0].copy())
buildinglines = clean_gdf(data[1].copy())
buildingparts = clean_gdf(data[2].copy())
exeter_area = clean_gdf(data[3].copy())

# Ensure all are in EPSG:4326 for folium
if buildings.crs != 'EPSG:4326':
    buildings = buildings.to_crs('EPSG:4326')
if buildinglines.crs != 'EPSG:4326':
    buildinglines = buildinglines.to_crs('EPSG:4326')
if buildingparts.crs != 'EPSG:4326':
    buildingparts = buildingparts.to_crs('EPSG:4326')
if exeter_area.crs != 'EPSG:4326':
    exeter_area = exeter_area.to_crs('EPSG:4326')

# Create 1km buffer around Exeter area
exeter_buffer = exeter_area.copy()
exeter_buffer['geometry'] = exeter_buffer.buffer(1000/111320)  # 1km in degrees approx

# Union buffer for spatial join
buffer_union = unary_union(exeter_buffer.geometry)

# Filter buildings within 1km buffer
buildings_within = buildings[buildings.intersects(buffer_union)]
buildinglines_within = buildinglines[buildinglines.intersects(buffer_union)]
buildingparts_within = buildingparts[buildingparts.intersects(buffer_union)]

# Clean filtered data again
buildings_within = clean_gdf(buildings_within)
buildinglines_within = clean_gdf(buildinglines_within)
buildingparts_within = clean_gdf(buildingparts_within)

# Create folium map centered on Exeter area
centroid = exeter_area.geometry.iloc[0].centroid
m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

# Plot Exeter area
folium.GeoJson(exeter_area.geometry.iloc[0], name='Exeter Area').add_to(m)
```

```python
    # Plot buffer
    folium.GeoJson(exeter_buffer.geometry.iloc[0], name='1km Buffer', style_function=lambda x: {'color': 'blue', 'fill': False, 'weight':2, 'dashArray':'5, 5'}
    # Plot buildings
    if not buildings_within.empty:
        folium.GeoJson(buildings_within[['geometry']], name='Buildings', style_function=lambda x: {'color': 'red', 'fill': True, 'fillOpacity':0.4}).add_to(m)
    # Plot building lines
    if not buildinglines_within.empty:
        folium.GeoJson(buildinglines_within[['geometry']], name='Building Lines', style_function=lambda x: {'color': 'green', 'weight':2}).add_to(m)
    # Plot building parts
    if not buildingparts_within.empty:
        folium.GeoJson(buildingparts_within[['geometry']], name='Building Parts', style_function=lambda x: {'color': 'orange', 'fill': True, 'fillOpacity':0.3}

    folium.LayerControl().add_to(m)

    # Save map
    map_filename = 'buildings_within_1km_exeter.html'
    m.save(map_filename)

    # Prepare summary
    summary = f"Buildings within 1km of Exeter area: {len(buildings_within)} polygons, {len(buildinglines_within)} lines, {len(buildingparts_within)} parts.\n"
    summary += f"First 5 building osids: {buildings_within['osid'].head(5).tolist()}\n"
    summary += f"First 5 buildingline osids: {buildinglines_within['osid'].head(5).tolist()}\n"
    summary += f"First 5 buildingpart osids: {buildingparts_within['osid'].head(5).tolist()}\n"

    return [summary, 'buildings_within_1km_exeter_map', 'Folium map showing all buildings, building lines, and building parts within 1km of the Exeter area pol
```

Buildings within 1km of Exeter area: 69169 polygons, 69342 lines, 85480 parts. First 5 building osids: ['32ff414d-d1c8-454b-9b67-ac472217e4ce', 'b9a49c80-fecf-4308-a98d-cf90ea4ea2e4', '9dfb4b41-f53d-44c0-8a7a-5f7884375677', 'aea058f2-6a39-4829-ae6b-85c6863a250e', 'ef3285ad-8a1d-4d4d-aa50-333888085e10'] First 5 buildingline osids: ['57202087-5453-473f-b927-0b58b9e564fb', 'd87c4727-0467-40c3-b148-1fae6bef930c', '38e4a6bd-c82d-4fe5-908f-df1497b7a199', '4ad601b7-2d43-4ffd-a813-785bdfac9349', '4fc5cd11-808d-4791-8676-b52213bce65b'] First 5 buildingpart osids: ['c7f949ce-e1f7-425d-b6fa-8ddc11065214', '7436b5c6-ff93-41cb-aeb1-c3c22acaf011', '6fcc76f4-817e-46e4-92c7-6bd50bd9c52d', '90474fb7-6565-4104-ba7b-21f7fe084140', 'cd808371-63d1-49b0-993e-87397a2aef3d']

A folium map has been generated showing all buildings, building lines, and building parts within 1km of the Exeter area polygon, with the buffer and area highlighted.

Map artifact: buildings_within_1km_exeter.html

host_agent

All buildings within 1km of Exeter have been identified. This includes:

- 69,169 building polygons
- 69,342 building lines
- 85,480 building parts

A map has been generated showing all these buildings, building lines, and building parts within 1km of the Exeter area polygon, with the buffer and area highlighted.

If you need to view or analyze the map, let me know!